

SMART INDIA HACKATHON 2025

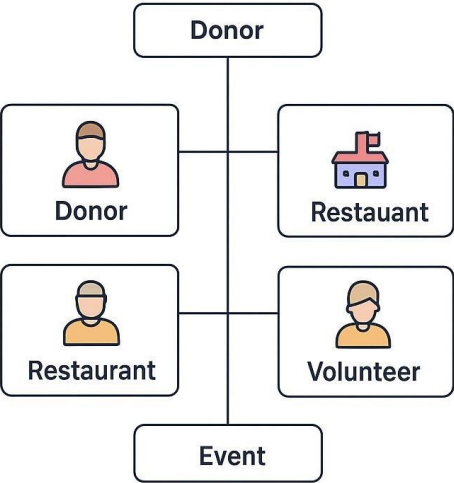


- **Problem Statement ID – SIH25060**
- **Problem Statement Title- Food Waste Management**
- **Theme- Clean & Green Technology**
- **PS Category- Software**
- **Team ID- 19**
- **Team Name - Team Learner**

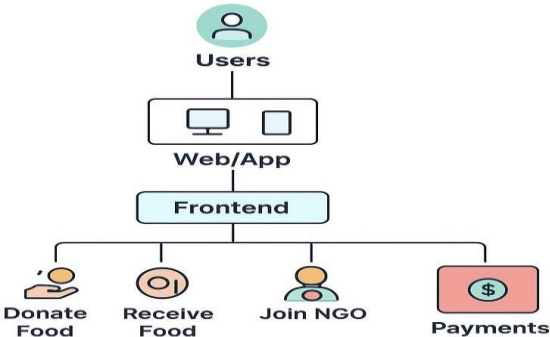


❖ Proposed Solution:-

- AI-powered Donation & Demand Prediction.**
 → Predicts food surplus and shortage areas using past donation patterns, event data & community demand.
- Smart Multi-channel Notifications.**
 → Updates via SMS, WhatsApp, Email & app alerts for NGOs, donors & receivers.
- Govt/NGO Database Integration.**
 → Links donations with government/NGO-verified shelters, food banks & beneficiaries.
- Emergency Case Prioritization.**
 → Flags urgent needs like disaster relief or child malnutrition for faster service.
- Dynamic Resource Allocation.**
 → Real-time dashboards for food collection centers, volunteers & delivery vehicles.
- Rural & Multi-language Support.**
 → Simple UI, regional languages & low-bandwidth mode for rural/remote users.



ER Diagram



System Architecture

TECHNOLOGY STACK :

- **Frontend : React.js, TailwindCSS, React Router**
- **Backend : Node.js + Express.js, Socket.IO, JWT**
- **Notifications : Firebase Cloud Messaging (FCM), Twilio**
- **Deployment : Netlify, Render**
- **Database Tools: MongoDB Compass**
- **Add-on (Upcoming Features) : AI Chatbot**

FEASIBILITY AND VIABILITY



Feasibility :

- Open-Source & Free Cloud** → Uses React, Node.js, MongoDB, Flask on free tiers (Netlify, Render, Atlas).
- Real-Time Food Updates** → Powered by Socket.IO for instant donor–NGO–receiver sync.
- Inclusive & Easy** → Simple UI + multi-language notifications for wider adoption (NGOs, donors, rural users).
- Low Cost & Scalable** → Minimal infra cost, expandable to multiple NGOs, cities & states.

Business Potential :

- Freemium Model** → Basic donation matching free, advanced analytics as paid.
- NGOs/Corporates** can subscribe → ₹1000–₹5000/month.
- Government/CSR adoption** → Large-scale contracts for Zero Hunger initiatives.

Viability :

- Scalability** → Works for local NGOs, food banks, disaster relief centers, and expandable to nationwide networks.
- Sustainability** → Low infra cost + reduced food wastage ensures long-term adoption.
- Social Impact** → Helps fight hunger, supports zero-waste goals, improves nutrition availability.
- Alignment with SDGs** → Supports *Zero Hunger (SDG 2)* & sustainability initiatives.

Challenges :

- Resistance from NGOs/partners to adopt new systems.
- Limited internet access in rural/remote areas.
- Food quality & safety assurance.
- Awareness gap among small donors/volunteers.
- Notification costs (SMS/WhatsApp) at large scale.

Future Monetization :

- Analytics dashboards for NGOs & corporates.
- Integration with restaurants, grocery chains & e-commerce.
- Logistics tie-ups with delivery partners.

IMPACT AND BENEFITS



Benefits of the solution :

Economic :

- Low-cost solution using free/open-source tech.
- Saves storage & transportation costs by reducing wastage.
- Scalable for NGOs, corporates, food banks & govt. programs with strong business potential.

Environmental :

- Waste Reduction** → Prevents food from ending up in landfills, cutting methane emissions.
- Reduced Travel & Overcrowding** → Optimized food delivery reduces unnecessary transport, lowering fuel use & carbon emissions.
- Efficient Resource Usage** → Better utilization of surplus food, reducing pressure on natural resources.

Educational :

- Promotes awareness about responsible food usage & waste reduction.
- Encourages NGOs, donors & citizens to adopt digital donation tools.
- Trains volunteers & staff in modern digital food distribution systems.

Impact on Audience :

- Beneficiaries** → Get timely access to safe & nutritious food, improving quality of life.
- NGOs & Volunteers** → Gain confidence in a transparent, efficient distribution system.
- Donors & Corporates** → See measurable social impact through analytics dashboards & CSR reporting.
- Judges & Evaluators** → Recognize innovation: AI + Real-time Matching + NGO/Govt. integration.
- Society & Government** → Supports India's "Zero Hunger" and "Clean & Green Technology" initiatives, boosting trust in digital solutions.

Technological :

- Real-time food surplus & demand matching with Socket.IO.
- AI-powered donation & demand prediction (Flask + ML).
- Multi-language notifications, cloud deployment, scalable architecture.

RESEARCH AND REFERENCES



References :

Alumni Platforms :

- **Feeding India** – <https://www.feedingindia.org>
- **No Food Waste** – <https://www.nofoodwaste.org>
- **ResQ Food App** – <https://resqapp.org>

Research :

- **AI for Food Supply Chain Optimization** – Springer Link
- **Reducing Food Wastage with Predictive Analytics** – IEEE Xplore
- **Digital Platforms for Food Redistribution** – ResearchGate

Feature	Existing Systems	Our Project
Online food donation booking	☒	☒
Walk-in donor/NGO support	☐	☒
Real-time food availability	☒ (limited)	☒
Multi-language notifications	☐	☒
AI-based demand prediction	☐	☒
Govt/NGO database integration	☐	☒
Free & open-source stack	☐ (licensed)	☒
Low-cost scalable deployment	☐	☒
Hybrid model (online + offline)	☐	☒